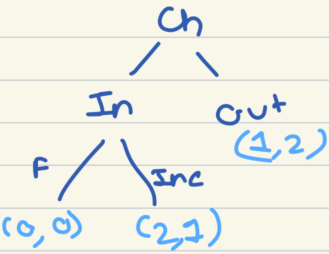
12/11

SPE: s^* is SPE if for each h, such that $P(h) = i$, $u_i(O_h(s^*)) \geq u_i(O_h(\gamma, s_{-i}^*))$ for all γ strategy of i

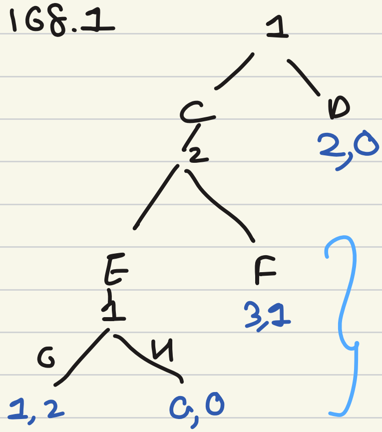


$s^* = (\text{Out}, F) \rightarrow$ Terminal history: Out

↳ NE, not SPE,

↳ perturbed steady state

Que - 168.1



NOT SPE

NE: (CH,F), (DG,E), (DH,E)

TH: (C,F)

$\Gamma(C,E)$

Finite horizon games & backward induction

Stage-1 take subgames of length: 1

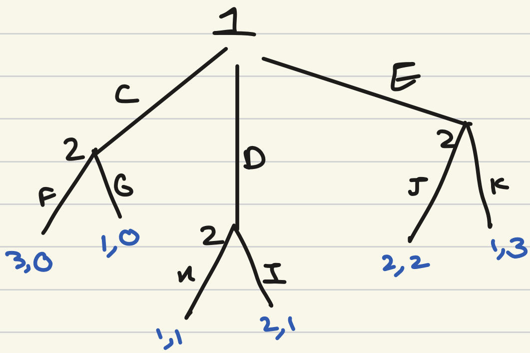
Hence: subgames of length: 1 in $\Gamma(C,E)$, $P(C,E)$: 1

Optimal action of 1 is G.

Stage - 2 Here sub games of length : 2 : $\Gamma(c)$
$P(c) = 3$, optimal action of 2 : E

Stage - 3

3, $F(\emptyset)$, $P(\emptyset)$
= D



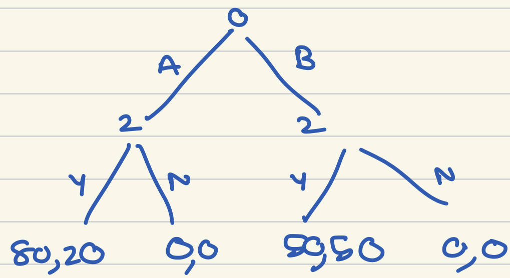
$\Gamma(C) = P(C) = 2$, optimal action : F & G
$\Gamma(D) = P(D) = 2$, " : H & I
$\Gamma(E) = P(E) = 2$, : K

optimal combinations of action (strategies)
FHK, FIK, GHK, GIK

$h = \phi, \; P(\phi) = 1$

with respect to FHK, optimal action of I = C
FIK, " = C
GHK, " = S, D, E
GIK, " = D

QUE - at the beggining of the course



SPE : (A, YY)
NE : (A, YY) (A, YN), (B, NY)

TH	Payoff
(A, Y)	(80, 20)

possible outcome =>

AY : 9
AN : 6
BY : 17
BN : 0

$$P_m = \frac{mve^2}{\left(R_e + r_L + r_2\left(\frac{1-s}{s}\right)\right)^2 + x^2} \cdot r_2\left(\frac{1-s}{s}\right)$$

why?
- Altruism
- Fear of revenge
- Spite, Reciprocity
- Sense of fairness